

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

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Applied Lab #1

An AI's Definition of AI

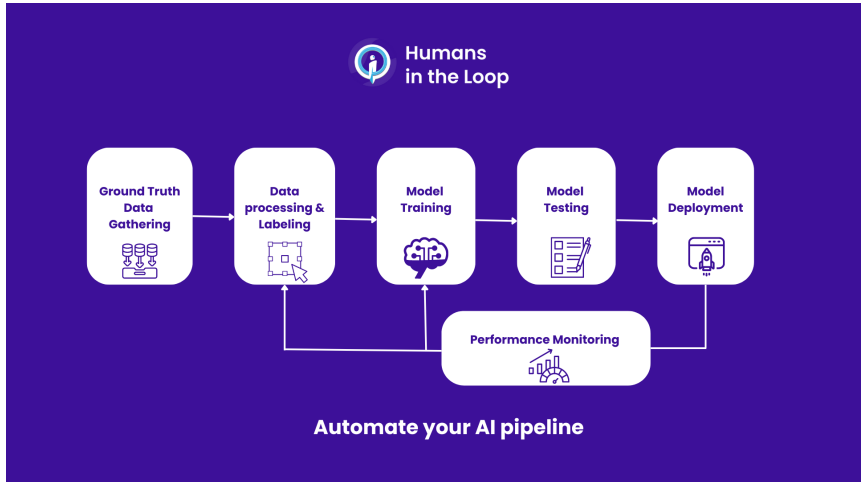
From Google Gemini:

“(AI) is a machine-based system that can, for a given set of human-defined objectives, emulate human cognitive functions – such as learning, reasoning, problem-solving, perception, and decision-making – to influence real or virtual environments.”

Key Points:

- Machine-Based
- Goal-Oriented
- Simulating Intelligence
- Autonomous / Adaptable
- External Impact

The AI “Pipeline”



(source)

Heikkilä and Arnett (2024):

- Circa 2010: curated, application-specific data
- 2017: Transformer models
 - + parallel processing = “foundation” models
 - More data → better performance
 - Data = WWW
- 2020s: Generative + multimodal models

Example: Common Crawl

[The Data](#) ▾[Resources](#) ▾[Community](#) ▾[About](#) ▾[Search](#) ▾[Contact Us](#)

Common Crawl maintains a
free, open repository of web
crawl data that can be used by
anyone.

Common Crawl is a 501(c)(3) non-profit founded in 2007.

We make wholesale extraction, transformation and analysis of
open web data accessible to researchers.

[Overview](#)

(Source: commoncrawl.org)

Implications (Heikkilä and Arnett 2024)

Effects:

- *Concentration* of Power
- Susceptibility to *Misinformation*
- Data *Provenance* is Poor / Nonexistent
- Cultural / Regional / etc. *Biases*

The Data Provenance Initiative

